PETROPHYSICS - LECTURE 10

Interactive activities student guide

From microcracks to macroscopic failure

Central question: how can tensile microcracks produced during compression organize into a macroscopic shear fault?

How to use this guide

- Work in pairs unless your instructor asks for individual predictions.
- Commit to a prediction before changing a slider.
- Change one control at a time and record the response.
- Separate observation, interpretation and mechanism.
- Use the reveal button only after writing your explanation.

Activity	Scale	Question
1. Flaw activation	One closed flaw	Which orientation slides first?
2. Wing-crack growth	One sliding flaw	Why does tensile cracking occur during compression?
3. Damage localization	Crack population	When does distributed damage become a fault?
4. Failure detective	Experimental evidence	Which deformation stage is best supported?

Name: _____ Date: _____

Activity 1 - Flaw activation laboratory

Learning goal: connect flaw orientation to resolved normal traction, shear traction and frictional activation.

Before touching the controls, rank beta = 15, 45 and 75 degrees from most likely to least likely to slide.

Prior knowledge from class

A closed flaw can slide only when the resolved shear traction exceeds its frictional resistance. Orientation changes both shear traction and normal clamping.

Step	What to do	What to notice
1	Keep the default stresses and friction. Test the three angle buttons.	Compare tau, sigma_n and the activation ratio.
2	Move beta slowly from 1 to 89 degrees.	Find the angle of maximum shear, then find the easiest frictional activation.
3	Increase friction while keeping the stresses fixed.	The activation ratio decreases and the optimum orientation shifts.
4	Increase confinement sigma_3.	Normal clamping and frictional resistance increase.
5	Use the Mohr circle to explain the selected stress point.	The point position must agree with the traction vectors on the flaw.

Equations used by the tool

$$\sigma_n = \frac{(\sigma_1 + \sigma_3)}{2} - \frac{(\sigma_1 - \sigma_3)\cos(2\beta)}{2}$$

$$\tau = \frac{(\sigma_1 - \sigma_3)\sin(2\beta)}{2}$$

Sliding criterion: $|\tau| \geq \mu \sigma_n$

beta	sigma_n	tau	|tau|/(mu sigma_n)	Locked or sliding?
15 deg				
45 deg				
75 deg				
Best angle				

Explain why the angle with maximum shear does not necessarily activate first when friction is present.

Activity 2 - Wing-crack growth laboratory

Learning goal: explain the sequence from parent-flaw sliding to tensile wing-crack nucleation, stable growth and arrest.

Hold sigma_1 fixed and predict how increasing sigma_3 changes sliding, nucleation and stable wing length.

Reading the two panels

- Crack geometry: the dark parent flaw slides in shear; orange wings open in tension.
- Opening-drive graph: K_I/K_Ic = 1 is the propagation threshold.
- The crossing back below 1 marks the stable arrest length.
- This is a semi-quantitative teaching model, not an experimental parameter inversion.

Step	What to do	What to notice
1	Start at low sigma_1 and increase it gradually.	Identify locked, sliding, nucleation and stable-growth states.
2	Increase the initial flaw size c.	A larger flaw raises the stress-intensity scale and promotes nucleation.
3	Increase K_Ic.	Greater toughness suppresses nucleation and shortens the wing.
4	Use the low/high confinement comparison.	Confinement delays sliding and adds a closing contribution on the wing.
5	Follow the wing direction during growth.	The trajectory approaches the sigma_1 direction as the wing lengthens.

Record the confinement comparison

Quantity	Low sigma_3	High sigma_3	Reason for the change
Parent flaw state			
K_I/K_Ic at l = 0			
Stable l/c			
Remote sigma_1 needed			

Resolve the paradox: how can a tensile crack grow while all remote principal stresses are compressive?

Activity 3 - Damage and localization simulator

Learning goal: distinguish distributed crack growth from spatially organized, connected damage.

Can many stable, noninteracting wing cracks create a system-spanning fault? Write your prediction before the first run.

Important limitation

This simulator uses schematic interaction rules. Use it to reason about crack connectivity and localization. Do not interpret its loading value as a predicted experimental failure stress.

Step	What to do	What to notice
1	Switch interaction OFF. Advance loading in steps.	Track active crack count, largest cluster and the macroscopic curve.
2	Continue to high loading with interaction off.	Notice whether isolated wings create a connected damage band.
3	Reset the population and switch interaction ON.	Repeat the same loading history for a controlled comparison.
4	Watch the preferred band and cluster fraction.	Linkage can increase sharply even when the total crack count changes gradually.
5	Compare ultrasonic paths A, B and C.	The path crossing the emerging band should show the largest velocity loss.

Record two controlled runs

Run	Active cracks	Largest cluster	Localization index	Fault-crossing Vp	Curve response
Interaction off					
Interaction on					

Which quantity provides the clearest evidence of localization? Explain why mean crack density alone is insufficient.

Activity 4 - Failure detective

Learning goal: build an interpretation progressively and separate observation, interpretation and mechanism.

After every evidence step, choose one stage and give a confidence score from 1 to 5. You may revise your answer, but identify the observation that changed it.

Recommended sequence

Step	What to do	What to notice
1	Select Case C and reveal only stress-strain.	Describe curve position and shape without naming a crack process.
2	Choose a stage and confidence.	A stress-strain curve alone may not uniquely identify localization.
3	Reveal volumetric strain.	Determine whether dilation or compaction adds evidence for damage.
4	Reveal ultrasonic paths.	Strong path dependence provides spatial information.
5	Reveal the fracture map and then the answer.	Imaging directly tests geometry, connectivity and localization.

Evidence log

Evidence available	Observation only	Chosen stage	Confidence 1-5	What changed?
Stress-strain				
+ Volumetric strain				
+ Ultrasonic paths				
+ Fracture map				

Final three-level explanation

Observation: What did the instruments record?

Interpretation: What deformation stage is best supported?

Mechanism: What physical crack process could produce the observations?

Synthesis - From one flaw to macroscopic failure

Use the four activities to reconstruct the complete causal sequence.

Scale	Physical process	Observable consequence
One closed flaw	Resolved shear overcomes friction.	Departure from purely elastic response may begin.
Flaw tip	Sliding generates a local tensile stress concentration.	A tensile wing nucleates when K_I reaches K_Ic.
Isolated wing	Wedging weakens and clamping grows with extension.	Stable growth and arrest are possible.
Crack population	Stress fields interact and cracks link.	Damage becomes spatially concentrated.
Specimen scale	A connected fault forms and begins to slide.	Peak stress, weakening and residual strength.

Final questions

1. Why is peak stress not the beginning of damage?

2. Why can an isolated wing crack not explain a macroscopic shear fault?

3. Which observation best distinguishes distributed damage from localization?

4. State the two distinct ways in which confinement suppresses brittle failure.

Key references

Paterson, M. S., and Wong, T.-F. (2005), Experimental Rock Deformation - The Brittle Field, Chapter 6.

Bernabe, Y., and Pec, M. (2022), Brittle creep and failure: a reformulation of the wing crack model.

Aben, F. M., Brantut, N., and Mitchell, T. M. (2020), Off-fault damage characterization during and after experimental rupture.

Renard et al. (2018) and McBeck et al. (2021), damage evolution and fracture-network localization.